Determining the Intrinsic properties of Porous Media

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***Abstract*—The concept of impedance through porous media holds significant importance in acoustical engineering. One important problem is optimizing airflow while minimizing noise. This paper explores the problem through the study of the acoustic properties of porous rigid media. The study examines the acoustic impedance through media with varying pore properties, focusing on the influence of varying pore numbers, percent porosity, and pore length on acoustic impedance. By maintaining the material as a constant, the scenario was simplified, enabling the identification of the relationship between pore properties and acoustic impedance. Previous studies done on porous media have mostly focused on how the material of the media affects the impedance. The results of this study expands the work done by previous studies, offering a new perspective on acoustic impedance. The porous media were created using Python to ensure the pores are uniformly spread across the cylinders. This experiment uses a traditional impedance tube with a speaker at the top of the tube and two microphones above the porous medium at the bottom. We will use frequencies ranging from one hundred fifty Hertz to one thousand Hertz. Analysis of reflected waves off the medium will provide the necessary impedance values.**

I. Introduction and Mathematical Theory

In acoustic engineering many porous materials used do not have controlled pore properties. This research uses a 3D printer to control the pore geometries of porous media, and then the intrinsic properties of the medium are determined. A relationship between the pore properties and characteristic impedance would allow for the creation of materials that more effectively impede specific frequencies of sound. The transfer function method is used for this analysis [1]. The complex transfer function H_{MC} is the microphone corrected transfer function which is defined as the ratio of the acoustic pressure measured at two microphone positions. The details for acquiring the corrected transfer function described in section II-C.

$$H_{MC} = \frac{P_{top}}{P_{bottom}} \tag{1}$$

From this transfer function, the surface impedance Z_s at the front face of the material is calculated. This relates the incident wave to the sample's resistance to the air particle velocity:

$$Z_s = j\rho c \frac{\sin(k(l-s)) - H_{MC}\sin(kl)}{H_{MC}\cos(kl) - \cos(k(l-s))} \tag{2}$$

where k is the wavenumber in air, l is the distance from the first microphone to the sample, and s is the spacing between microphones. Because the sample is backed by a rigid-walled air cavity of depth D, we must account for the backing impedance Z_b, which is given by:

$$Z_b = -jZ_{air}\cot(kD) \tag{3}$$

To isolate the intrinsic properties from the system effects, the "Two-Cavity Method" is employed. A second set of measurements is taken using a different air backing space, and all variables associated with this second configuration are denoted with a prime ($'$). This allows for the calculation of the characteristic impedance Z_c:

$$Z_c = \sqrt{\frac{Z_s Z_s'(Z_b - Z_b') - Z_b Z_b'(Z_s - Z_s')}{Z_s - Z_s' - (Z_b - Z_b')}}. \tag{4}$$

The characteristic impedance (Z_c) physically represents the amount of energy that is reflected off of the surface of the medium. The propagation constant γ can also be calculated with these values:

$$\gamma = \frac{1}{2L}\ln\left(\frac{Z_s + Z_c}{Z_s - Z_c}\cdot\frac{Z_b - Z_c}{Z_b + Z_c}\right) \tag{5}$$

The propagation constant (γ) physically represents the exponential loss of pressure within the medium. These equations are outlined by Utsuno et al. [1] which provides the framework for Z_c and γ.

A. Derivation of Internal Wave Coefficients

Following the mathematics of Hou and Bolton [2] for plane waves in a standing wave tube, we can apply the boundary conditions of continuity for pressure and particle velocity. Their math can be used to solve for the internal forward and backward traveling wave coefficients (A and B) inside the porous medium.

Inside the material, the pressure $P_m(x)$ and particle velocity $V_m(x)$ are governed by the material's intrinsic properties, Z_c and γ, and can be expressed as:

$$P_m(x) = Ae^{-\gamma x} + Be^{\gamma x} \tag{6}$$

$$V_m(x) = \frac{A}{Z_c}e^{-\gamma x} - \frac{B}{Z_c}e^{\gamma x} \tag{7}$$

At the front interface of the sample ($x = 0$), let P_0 and V_0 represent the known total pressure and velocity. Evaluating the internal equations at $x = 0$ yields a system of two equations:

$$P_0 = A + B \tag{8}$$

$$V_0 = \frac{A - B}{Z_c} \tag{9}$$

Multiplying the velocity equation by Z_c gives $A - B = Z_c V_0$. This system of equations can be solved for the coefficients A and B:

$$A = \frac{1}{2}(P_0 + Z_c V_0) \tag{10}$$

$$B = \frac{1}{2}(P_0 - Z_c V_0) \tag{11}$$

II. Methodology

The experiment was conducted using a standard impedance tube setup, closely following the methodology outlined in Utsuno's paper. This experiment differs primarily in its application: it is designed to measure the acoustic properties of 3D-printed porous media rather than natural materials such as sand. To start off a given experiment, a porous medium with known properties of pore length, pore shape, and pore size are placed in the impedance tube.

A. Experimental Configuration

The function generator seen in Fig. 1 provides a signal to the speaker at one end of the impedance tube. A Python script controls the function generator, sweeping through frequencies from 150 Hz to 1,000 Hz in 10 Hz increments. At each frequency, the resulting sound wave is measured by two microphones. These microphones are connected to two differential preamplifiers to boost the signals before they are visualized and recorded on an oscilloscope. The oscilloscope captures the time-domain pressure signals, $p_1(t)$ and $p_2(t)$, which are then converted to the frequency domain to determine the complex pressures, P_1 and P_2, at each frequency. These complex pressures are required for Equation 1. To reduce measurement error, 16 oscilloscope averages are performed for each individual frequency. Additionally, for the oscilloscope to accurately distinguish between the 10 Hz frequency increments, the frequency resolution (df) must be smaller than 10 Hz. The frequency resolution is calculated using the following equation:

$$df = \frac{\text{sample rate}}{\text{number of samples}} \tag{12}$$

The sampling rate on the oscilloscope is set to 50,000 samples/s, and the sample size is set to 10,000 samples. This yields a frequency resolution of 5 Hz, ensuring that when the time-domain signal is converted to the frequency domain, the resolution is enough to correctly identify between 10 Hz increments.

An important component of this setup, highlighted by the blue region at the bottom of the impedance tube in Fig. 1, is the interchangeable air backing ring. This is the ring used in the two cavity procedure.

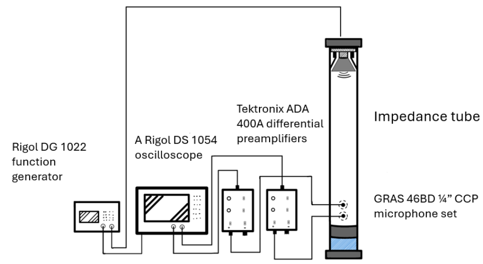

Fig. 1. Experimental Setup

B. Two-Cavity Procedure

A critical step in this method is varying the air backing depth. After the first frequency sweep is completed for an initial depth D, the air space ring at the back of the tube is replaced with a ring of a different depth, D'. As shown in Fig. 2, this creates two unique acoustic environments within the tube.

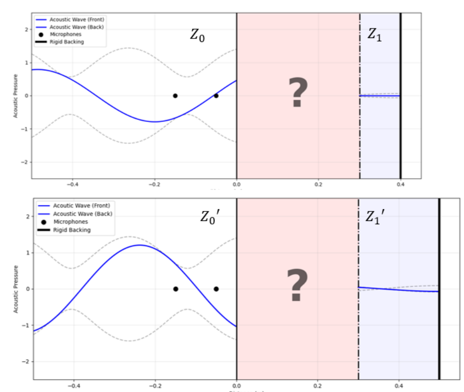

Fig. 2. Two Unique Acoustic Environments

This change in boundary conditions provides the modified impedance values, Z_s' and Z_b', which are necessary to calculate the characteristic impedance and propagation constant described by Equations 4 and 5.

C. Microphone correction

To account for and remove the inherent frequency response biases of the microphones, two frequency sweeps are performed for each air backing configuration. The first sweep is performed with the microphones in their standard configuration, giving a transfer function of:

$$H_{12} = \frac{\epsilon_1 P_{top}}{\epsilon_2 P_{bottom}} \tag{13}$$

where ϵ_1 and ϵ_2 are the frequency responses associated with microphone 1 and microphone 2, respectively.The second measurement is performed with the microphone positions swapped. This results in a second transfer function:

$$H_{21} = \frac{\epsilon_2 P_{top}}{\epsilon_1 P_{bottom}} \tag{14}$$

Combining these two frequency sweeps allows for the algebraic cancellation of the ϵ_1 and ϵ_2 terms, yielding the corrected transfer function, H_{MC}:

$$H_{MC} = \sqrt{H_{12} \cdot H_{21}} = \frac{P_{top}}{P_{bottom}} \tag{15}$$

III. RESULTS

A. Microphone Data

During the collection process the data from the microphones is actively plotted to ensure the data is being place in the correct frequency bin. Fig 3. is the data collected with mic 1 on top and mic 2 on bottom. When the microphones have been swapped, the data is again plotted. To verify the reproducibility of our data we can compare mic 1 of Fig. 3 to mic 2 of Fig. 4. These two paths should look similar since they are detecting the pressure at the same location and the same acoustic environment.

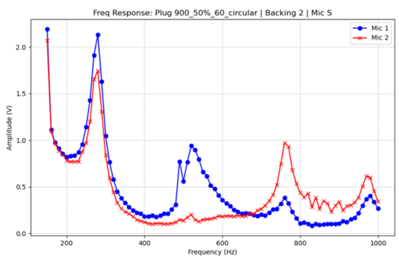

Fig. 3. Microphone Data with Original Microphone Configuration

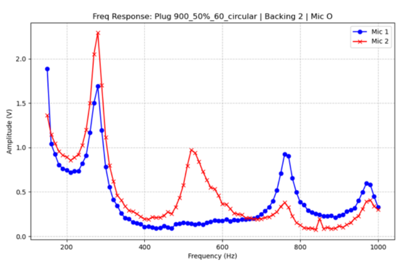

Fig. 4. Microphone Data with Swapped Microphone Configuration

B. Intrinsic Properties

Once the microphone data has been collected. The transfer function can be used to find the values of the propagation constant (γ) and the characteristic impedance (Z_c). Those values are plotted in Fig. 5.

This figure shows the characteristic impedance and propagation constant for a 900 pore 50% porous medium with 60 mm pores. Three seperate runs are plotted on the same figure. The intrinsic properties are complex numbers, so the imaginary part of each is reflected over the x-axis. This medium interestingly shows a significant increase in acoustic impedance from 500 to 600 Hz. This result was reproducible from run to run. The results from a 20 % porous medium with 900 pores and a length of 90 millimeter pores were also plotted in Fig. 6.

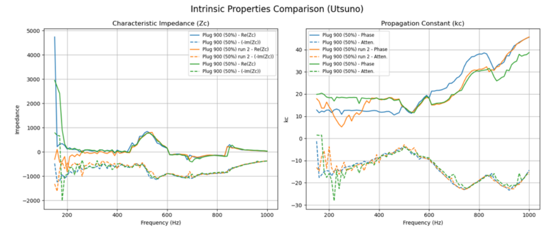

Fig. 5. Intrinsic Properties of 900 60 Millimeter Pores with 50% Porosity

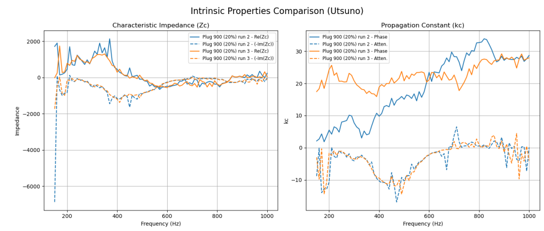

Fig. 6. Intrinsic Properties of 900 90 Millimeter Pores with 20% Porosity

This figure shows a noticeable characteristic impedance spike in the lower frequency range. This impedance spike is roughly 300 (Hz) lower than Fig. 5. In both figures, the lower frequencies have noticeably more variation in the data. This is attributed to the small surface area of the speaker not being able to push the required volumes of air for lower frequencies.

C. Visualization of intrinsic properties

The data plotted in Fig. 5 and 6 have been used to create a model that visualizes wave propagation through three distinct regions: incident air, porous sample, and back cavity. The equations outlined in section I-A are used to animate the wave in the tube. This visual shows the evolution of the acoustic environment for each individual frequency.

Fig. 7 shows the acoustic environment created by the data in Fig. 5.

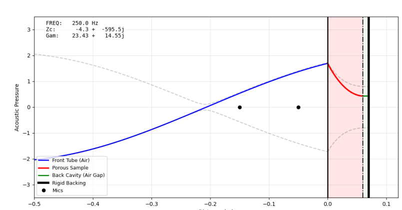

Fig. 7. Intrinsic Properties of the 60 Millimeter 50% Porosity Medium

Now the data from Fig. 6 is also shown in Fig. 8. These models demonstrate the increased attenuation through the 90 millimeter pug when compared to the 60 millimeter, which is to be expected. This model gives a physical intuition of the experimental data, and allows for the easy identification of non-physical data

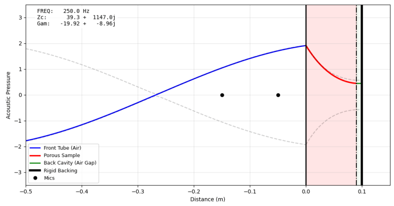

Fig. 8. Intrinsic Properties of 90 Millimeter 20% porous medium modeled

IV. Conclusion

This paper analyzes the intrinsic properties of porous media using the transfer function method. In this experiment, two plugs with differing pore geometries were tested. A noticeable increase in the characteristic impedance was found at specific frequencies. More plugs with differing pore properties will need to be analyzed to determine a relationship between this impedance spike and the properties of the medium. Characterization of this relationship could have significant benefits in acoustical engineering. Lastly, these experimental values were used in the creation of a model to better understand and communicate the data gathered. Future work on this research will increase the low frequency range to improve the accuracy of the data. Once a large enough dataset has been collected, a machine learning model trained on this data would be beneficial in overcoming the long data-collection times.

References

[1] H. Utsuno, T. Tanaka, T. Fujikawa, and A. F. Seybert, "Transfer function method for measuring characteristic impedance and propagation constant of porous materials," *Journal of the Acoustical Society of America*, vol. 86, no. 2, pp. 637–643, 1989.

[2] K. Hou and J. S. Bolton, "A transfer matrix method for estimating the dispersion and attenuation of plane waves in a standing wave tube," Publications of the Ray W. Herrick Laboratories, Purdue University, Tech. Rep. Paper 54, 2009. [Online]. Available: http://docs.lib.purdue.edu/herrick/54